

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMOE605A - QUANTITATIVE TECHNIQUES AND PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 In graphical solution of solving Linear Programming problem to convert inequalities into equations, we
 - a. Use Slack variables
 - b. Use Surplus variables,
 - c. Use Artificial surplus variables
 - d. Simply assume them to be equations
- 2 A variable which has no physical meaning, but is used to obtain an initial basic feasible solution to the linear programming problem is referred to as:
 - a. Basic variable
 - b. Non-basic variable
 - c. Artificial variable
 - d. None of the above
- 3 In assignment problem of maximization, the objective is to maximise
 - a. Profit
 - b. Optimization
 - c. Cost
 - d. None of the above
- 4 Which of the following is a method for improving an initial solution in a transportation problem?
 - a. northwest-corner
 - b. intuitive lowest-cost
 - c. southeast-corner rule
 - d. stepping-stone
- 5 A solution is not a basic feasible solution in a transportation problem if after allocations
 - a. there is no closed loop
 - b. there is a closed loop
 - c. there is degeneracy
 - d. total allocations $< m+n-1$

- 6 The purpose of a dummy row or column in an assignment problem is to
 - a. Obtain balance between total activities & total resources
 - b. Prevent a solution from becoming degenerate
 - c. Provide a means of representing a dummy problem
 - d. None of the above
- 7 If a customer enters in the queue but after some time loses patience and decided to leave then he is said to have
 - a. Balking
 - b. Jockeying
 - c. Reneging
 - d. Alternating
- 8 Mean time between two successive arrivals is equal to
 - a. Arrival Rate
 - b. Service Rate
 - c. Reciprocal of arrival rate
 - d. Reciprocal of service rate
- 9 'A' class materials consumes
 - a. 10 % of the total annual inventory cost
 - b. 30% of the total annual inventory cost
 - c. 70 to 75% of the total inventory cost
 - d. 90% of the total annual inventory cost
- 10 Which of the following increases with the quantity ordered per order?
 - a. Carrying cost
 - b. Ordering cost
 - c. Purchase cost,
 - d. Demand
- 11 PERT analysis is based on
 - a. Optimistic Time
 - b. Pessimistic Time
 - c. Most likely Time
 - d. All of the above
- 12 What does a crashing imply?
 - a. completing the project
 - b. abandoning the project
 - c. reducing the cost of project
 - d. reducing time for few of the activities

Q2. Solve the following.

- A) A firm manufactures two products A and B with a profit ₹4 and ₹5 respectively. The 6 products are made by the machines X and Y. The available machine time in minutes for the machines X and Y are 2000 and 1500 respectively. The time taken to make the product A by machines X and Y are 5 and 4 minutes and for the product B are 3 and 4 minutes respectively. Formulate the LP model for maximization of the profit.
- B) A company manufactures two products, X and Y by using three machines A, B, and C. 6 Machine A has 4 hours of capacity available during the coming week. Similarly, the available capacity of machines B and C during the coming week is 24 hours and 35 hours respectively. One unit of product X requires one hour of Machine A, 3 hours of machine B and 10 hours of machine C. Similarly one unit of product Y requires 1 hour, 8 hour and 7 hours of machine A, B and C respectively. When one unit of X is sold in the market, it yields a profit of Rs. 5/- per product and that of Y is Rs. 7/- per unit. Solve the problem by using graphical method to find the optimal product mix.

Q3. Solve the following.

- A) A department head has Five persons and Five Jobs have to be performed. 6
Persons differ in efficiency and jobs differ in their intrinsic difficulty. Time each man would take to perform each job is given in the effectiveness matrix. How the jobs should be allocated to each person so as to minimize the total man hours?

Person	Job				
	1	2	3	4	5
A	8	4	2	6	1
B	0	9	5	5	4
C	3	8	9	2	6
D	4	3	1	0	3
E	9	5	8	9	5

- B) Explain the North-West corner rule of Transportation Problems using a simple example. 6

Q4. Solve Any Two of the following.

- A) A branch of a Nationalized bank has only one typist. Since typing work varies in length 6 (number of pages to be typed), the typing rate is randomly distributed approximating a Poisson distribution with a mean service rate of 8 letters per hour. The letter arrives at a rate of 5 per hour during the entire 8-hour workday. If the typist is valued at Rs. 1.50 per hour, determine: (a) Equipment utilization, (b) The percent time an arriving letter has to wait, (c) Average system time
- B) In case of inventory system, calculate the Ordering Level, Minimum Level and Maximum Level from the following data: Re-order quantity 1,500 units Re-order period 4 to 6 weeks Maximum consumption 400 uts per week Average consumption 300 units per week Minimum consumption 250 units per week.
- C) What are the reasons for replacement of an item during its life? What are the types of 6 replacement problems?

Q5. Solve Any Two of the following.

- A) A manufacturer uses Rs. 10,000 worth of an item during the year. He has estimated the 6 ordering costs as Rs.25 per order and carrying costs as 12.5% of average inventory value. Find the optimal order size, number of orders per year, time period per order and total cost.
- B) Explain p- System and q - system of ordering material, which one you prefer? Give your 6 reasons.
- C) A taxi owner estimates from his past records that the costs per year for operating taxi 6 whose purchase price when new is Rs.60000/- are as given below:
After 5 years, the operating cost is Rs. $6000 \times k$ Where $k = 6, 7, 8, 9, 10$, i.e. 'k' denotes years. If the resale value decreases by 10% of purchase price each year, what is the best replacement policy? Cost of money is zero.

Age (Yrs)	1	2	3	4	5
Operating Cost, Rs.	10000	12000	15000	18000	20000

Q6. Solve Any Two of the following.

- A) Compare and contrast CPM and PERT. Under what conditions would you recommend 6 Scheduling by PERT? Justify your answer with reasons.

- B) What is meant by Project Crashing? A project has activities with the following normal and crash times and cost. Draw the network and Determine the cost slope and mention which activities need to be crashed first. 6

Activity	Predecessor activity	Normal time, Weeks	Crash Time, Weeks	Normal Cost, Rs.	Crash cost, Rs.
A	---	4	3	8000	9000
B	A	5	3	16000	20000
C	A	4	3	12000	13000
D	B	6	5	34000	35000
E	C	6	4	42000	44000
F	D	5	4	16000	16000
G	E	7	4	66000	72000
H	G	4	3	2000	5000

- C) A project schedule has the following characteristics. 6
- (a) Draw the project network and trace all the possible paths from it. (b) Determine the critical path and the expected project time. (c) What is the probability that the project will be completed in 20 weeks?

Activity	Expected Duration (Weeks)		
	Optimistic	Most Likely	Pessimistic
1-2	3	3	3
2-3	3	6	9
2-4	2	4	6
3-5	4	6	8
4-6	4	6	8
5-6	0	0	0
5-7	3	4	5
6-7	2	5	8

*** End ***